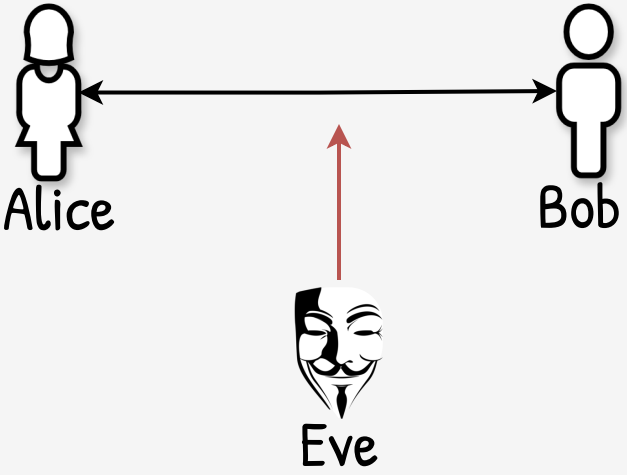
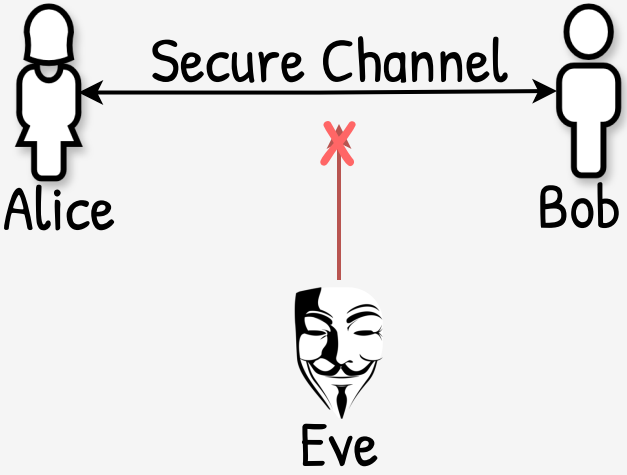
Cryptography I:
Stream Cipher, KDF, PRNG, MAC and Secure channels

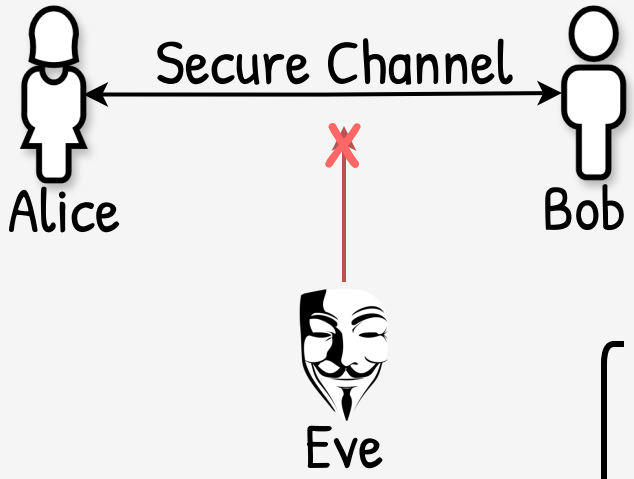
Taller de Seguridad

Ing. Martin Di Paola - Fiuba

Alice
Bob

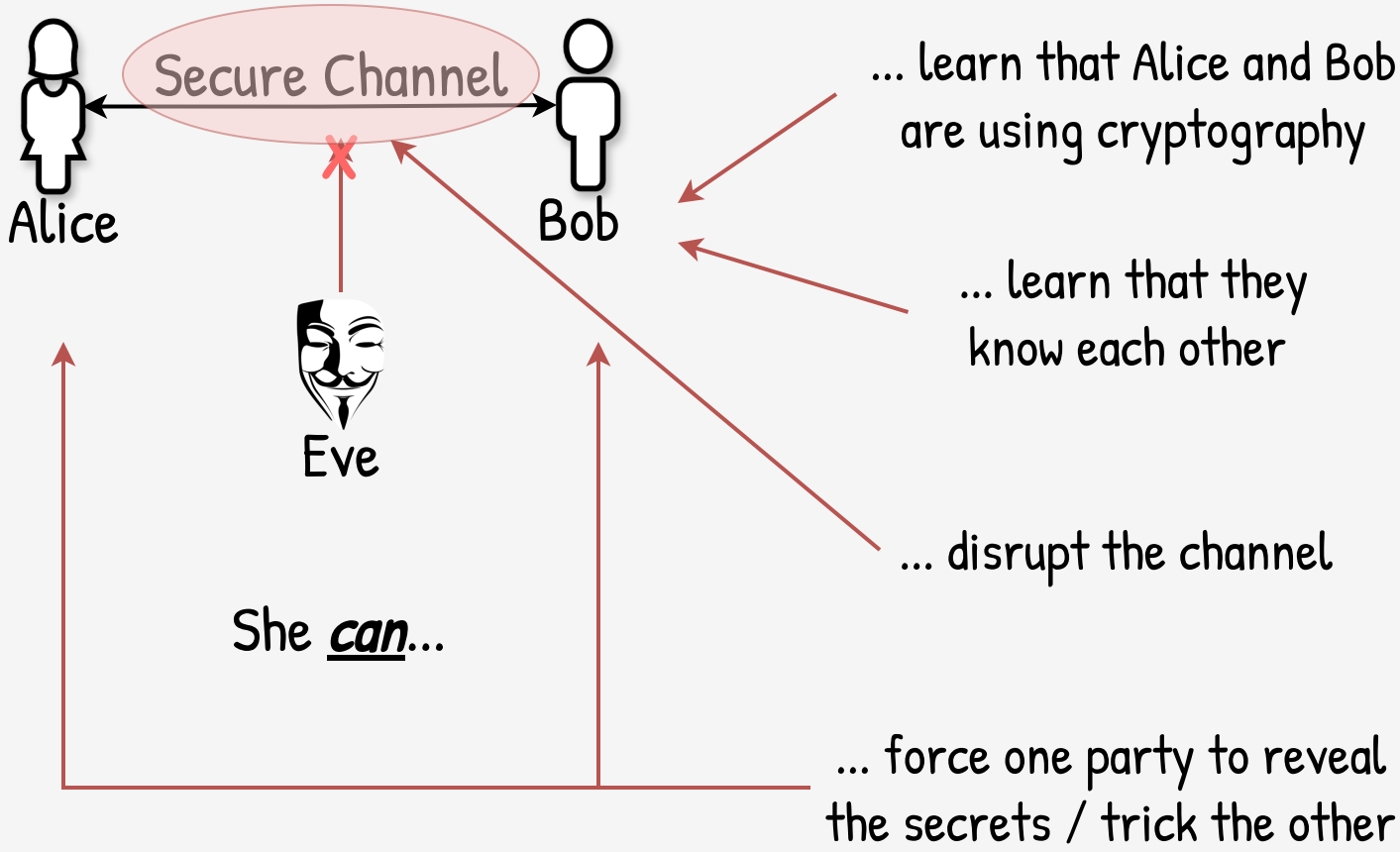

Alice
Bob
Eve


Secure Channel
Alice
Bob
Eve



She ***cannot***...

- ... learn anything about the msg — Confidentiality
- ... modify it without being noted — Integrity
- ... send fake msgs — Authentication


Secure Channel
Alice
Bob
Eve
... learn that Alice and Bob are using cryptography
... learn that they know each other
... disrupt the channel
She can...
... force one party to reveal the secrets / trick the other

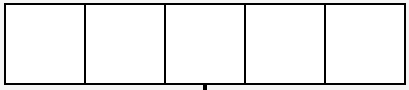
One Time Pad

ptext p

$+$

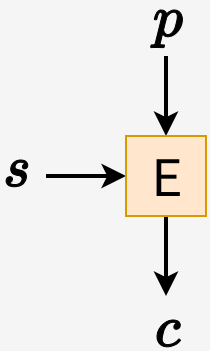
key stream s

ctext c

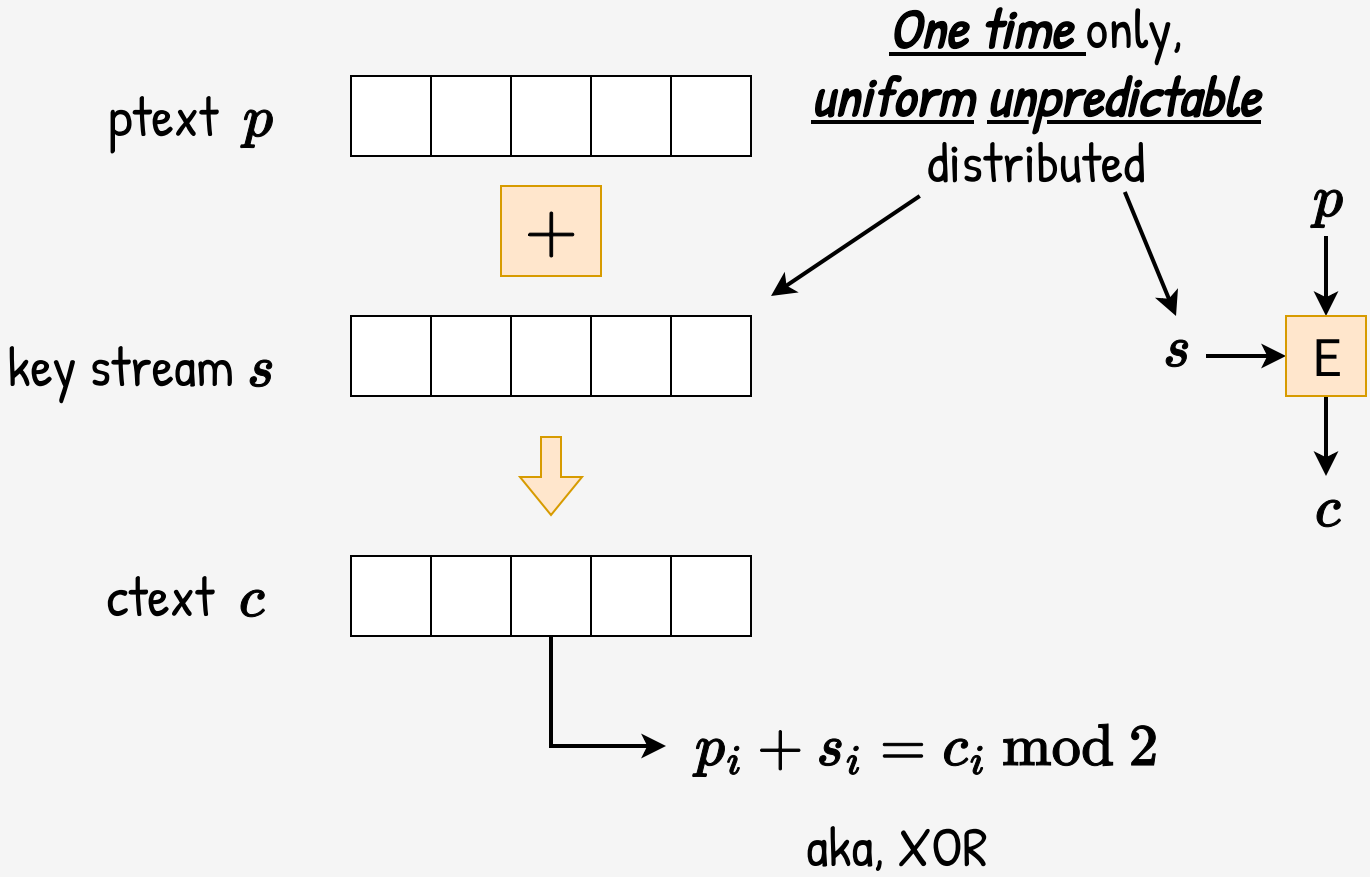


$$p_i + s_i = c_i \bmod 2$$

aka, XOR



One Time Pad


ptext p
+
key stream s
ctext c
$p_i + s_i = c_i \bmod 2$
aka, XOR
One time only,
uniform ***unpredictable***
distributed
p
s
E
c

One Time Pad

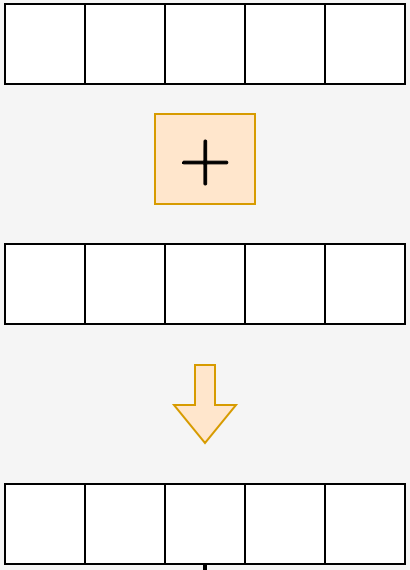
OTP gives confidentiality

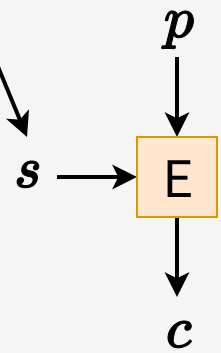
One time only,
uniform ***unpredictable***
distributed

ptext p

key stream s

ctext c





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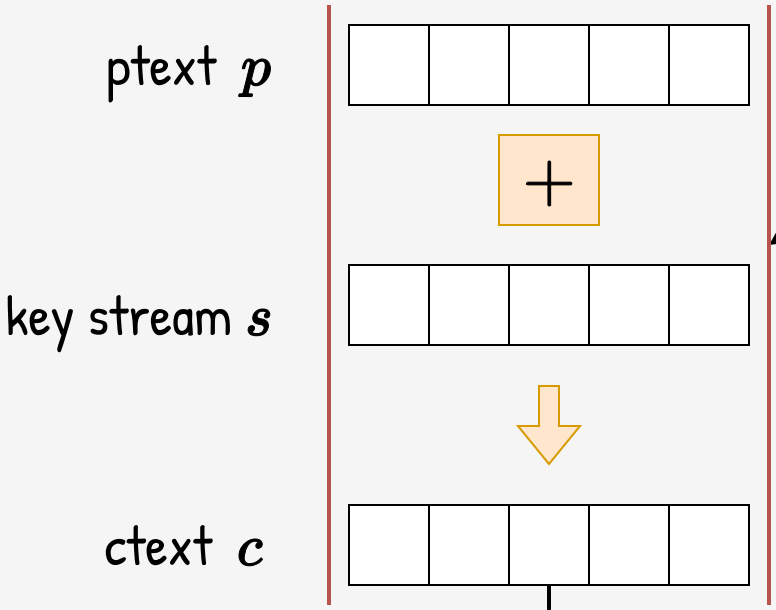
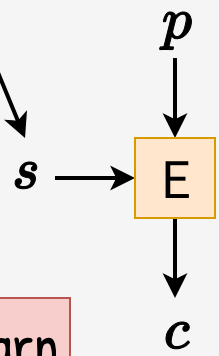
aka, XOR

One Time Pad



OTP gives confidentiality

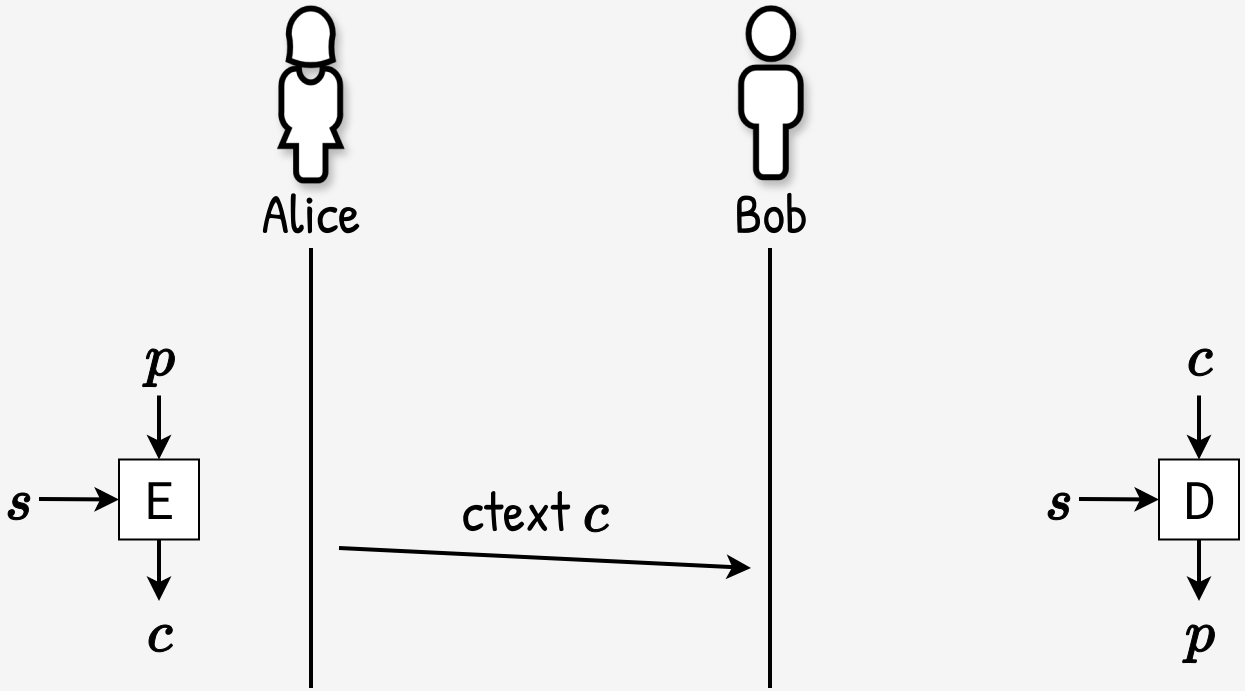
One time only,
uniform ***unpredictable*** distributed



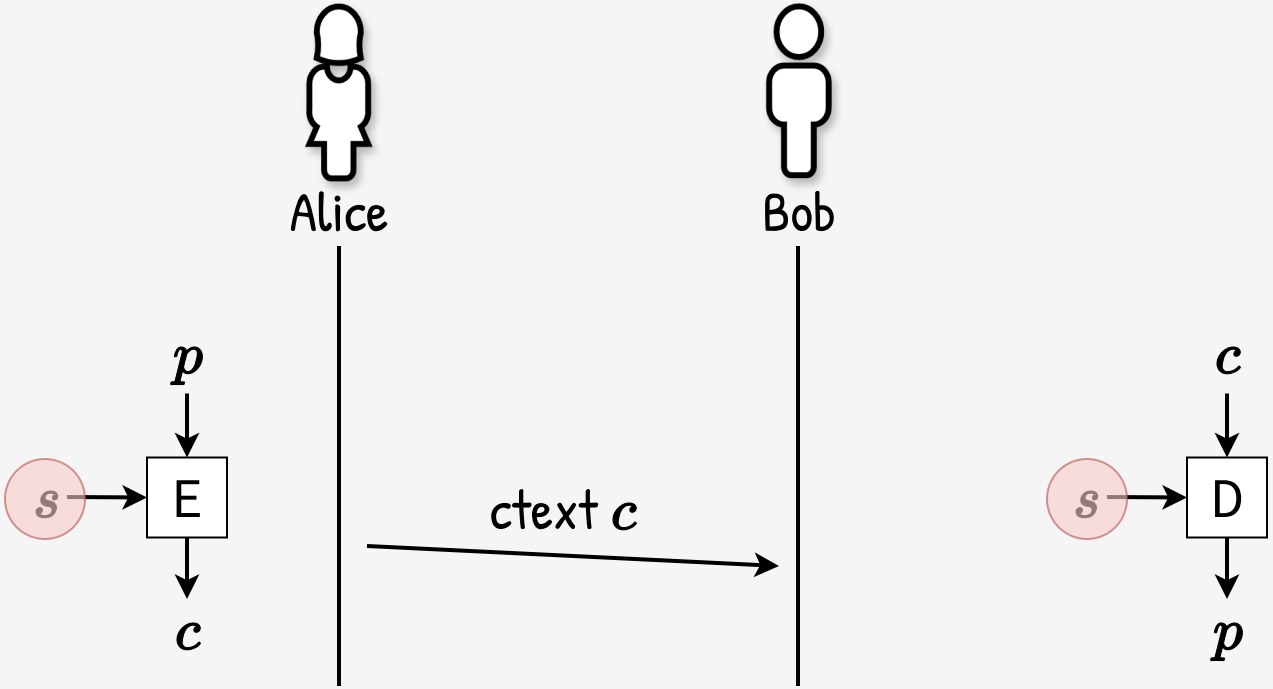
Attacker can learn the length of p

$$p_i + s_i = c_i \bmod 2$$

aka, XOR



Stream Cipher



How Alice can ***share*** with Bob the key stream s?

Stream Cipher

$$k \longrightarrow \boxed{G} \longrightarrow s$$

must be ***uniformly*** random and ***secret***

deterministic function

uniform unpredictable distributed, properties hold for a ***very large*** output ***length***

$s \leftarrow U$: **any** bit pattern is as likely as any other

Given $s_0, s_1, \ldots s_i$ it is impossible to predict **any** future s_j

$k \rightarrow \boxed{G} \rightarrow s$

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$k \rightarrow$ G $\rightarrow s$

must be ***uniformly*** random and ***secret***

deterministic function

uniform ***unpredictable*** distributed, properties hold for a ***very large*** output ***length***

Given s is impossible to leak **anything** about k

$s \leftarrow U$: **any** bit pattern is as likely as any other

Given $s_0, s_1, \ldots s_i$ it is impossible to predict **any** future s_j

$k \rightarrow$ G $\rightarrow s$

must be ***uniformly*** random and ***secret***

deterministic function

uniform ***unpredictable*** distributed, properties hold for a ***very large*** output ***length***

Given s is impossible to leak **anything** about k

Cryptographic Secure Pseudo Random (Number) Generator

chosen ***randomly***, but can be ***public*** as long ***it isn't repeated***

nonce n

$k \rightarrow$ G $\rightarrow s$

must be ***uniformly*** random and ***secret***

deterministic

function

Given s is impossible to leak **anything** about k

$s \leftarrow U$: **any** bit pattern is as likely as any other

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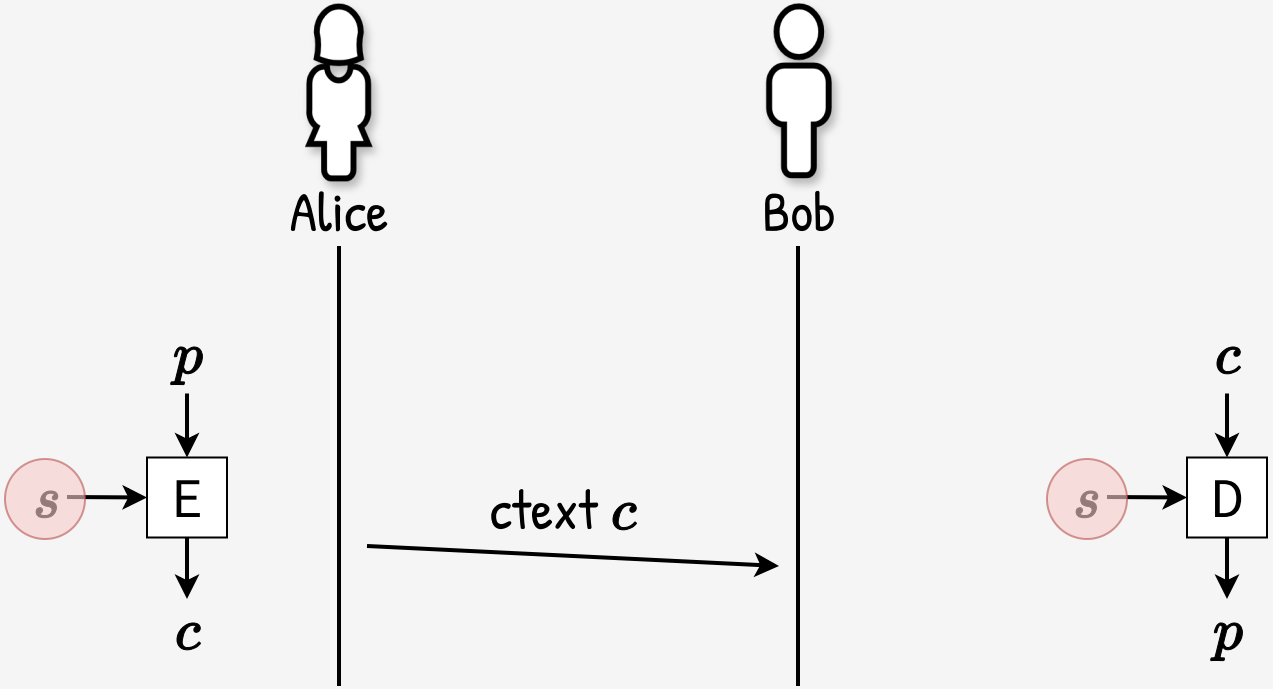
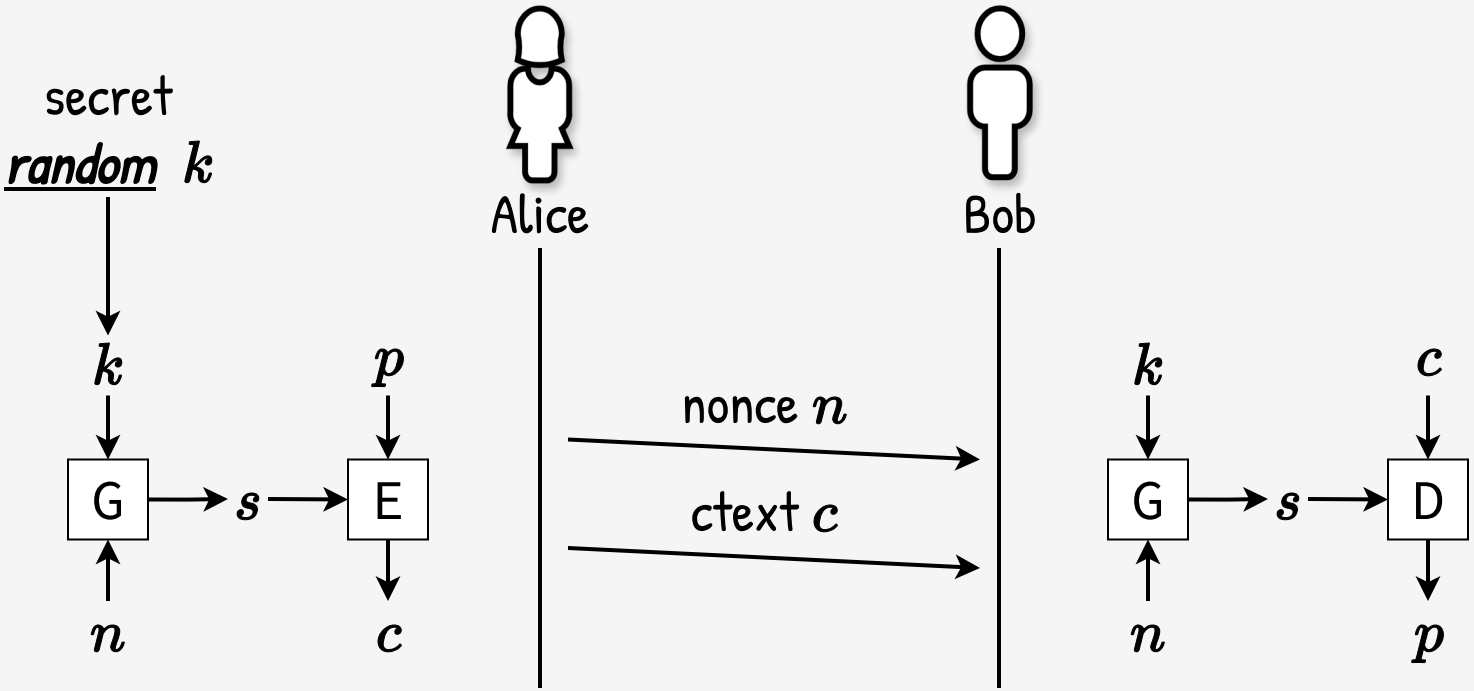
Cryptographic Secure

Pseudo Random (Number) Generator

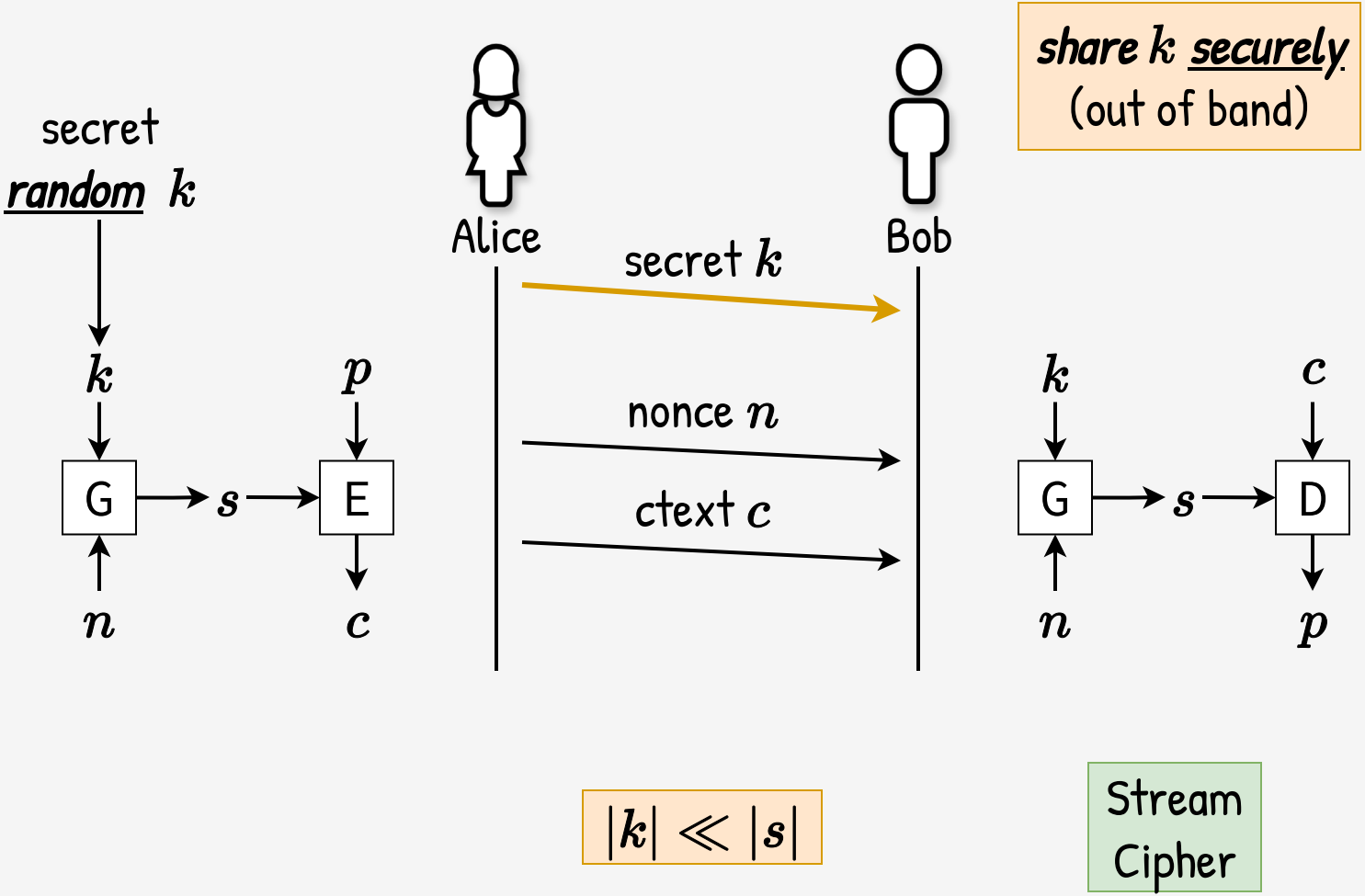


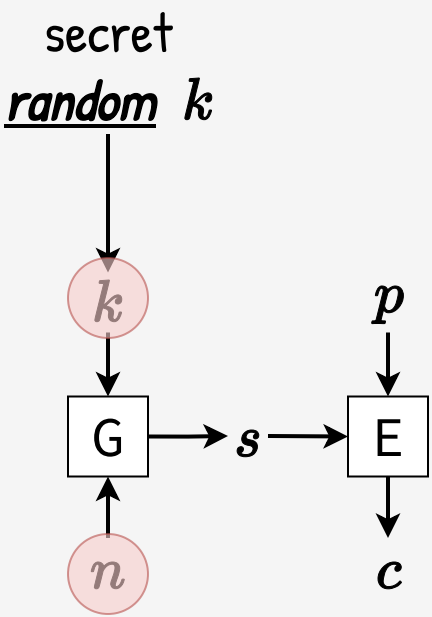
How Alice can ***share*** with Bob the key stream s?

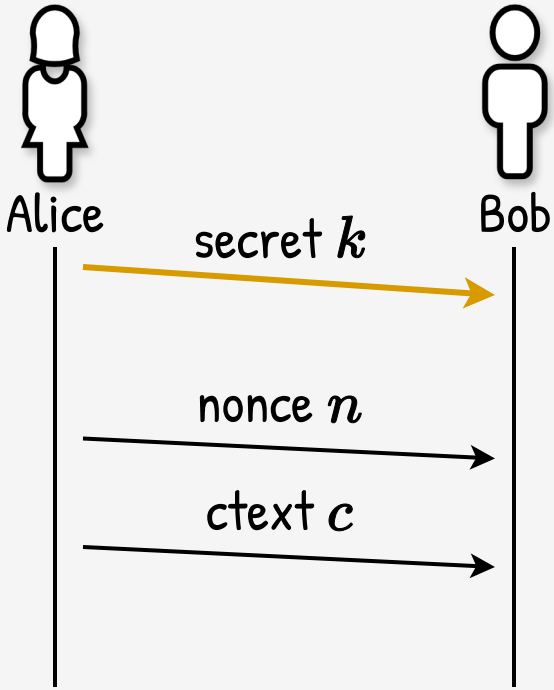
Stream Cipher

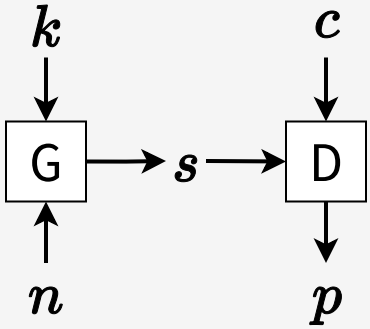


Stream Cipher


share k securely
(out of band)
secret
random k
k
p
G
s
E
n
c
Alice
Bob
secret k
nonce n
ctext c
k
c
G
s
D
n
p
$|k| \ll |s|$
Stream
Cipher





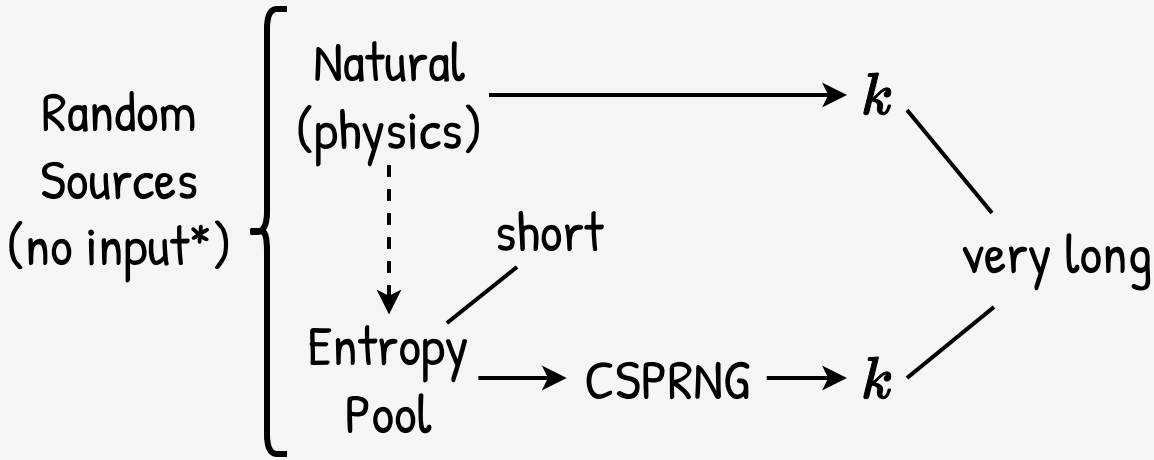


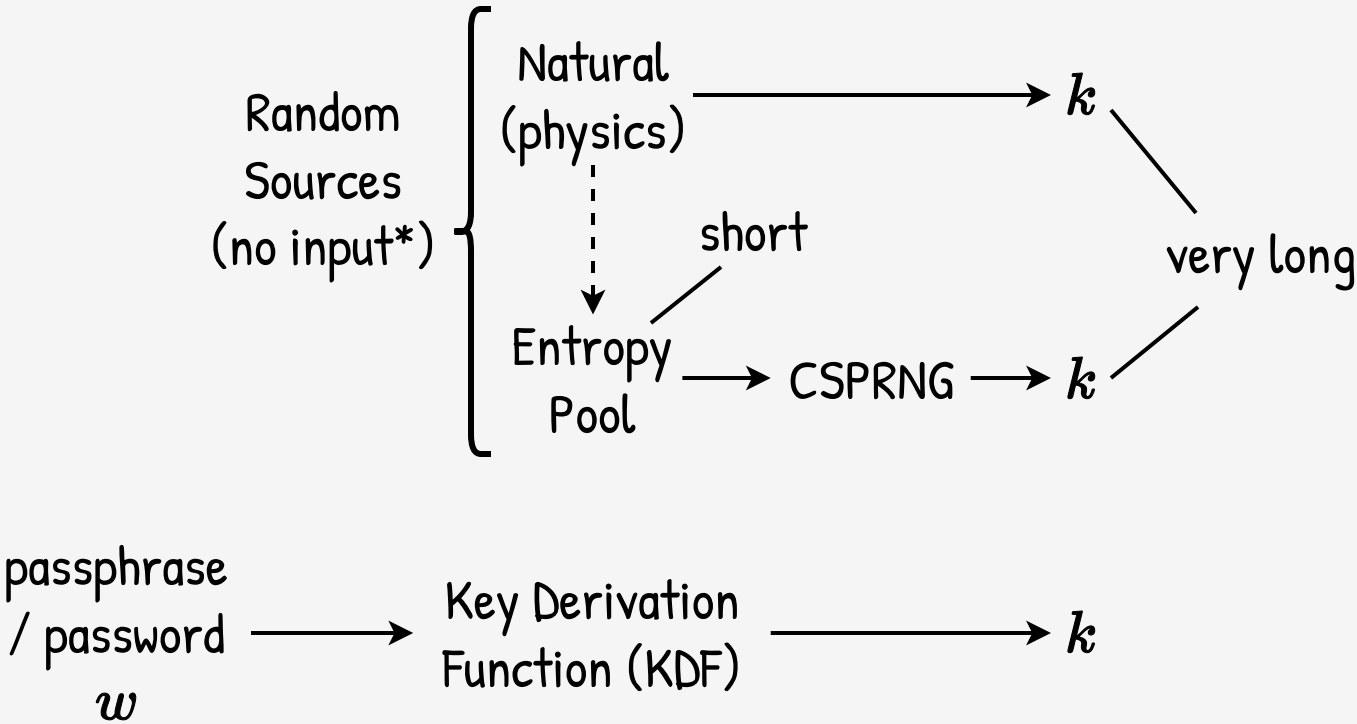
share k securely
(out of band)

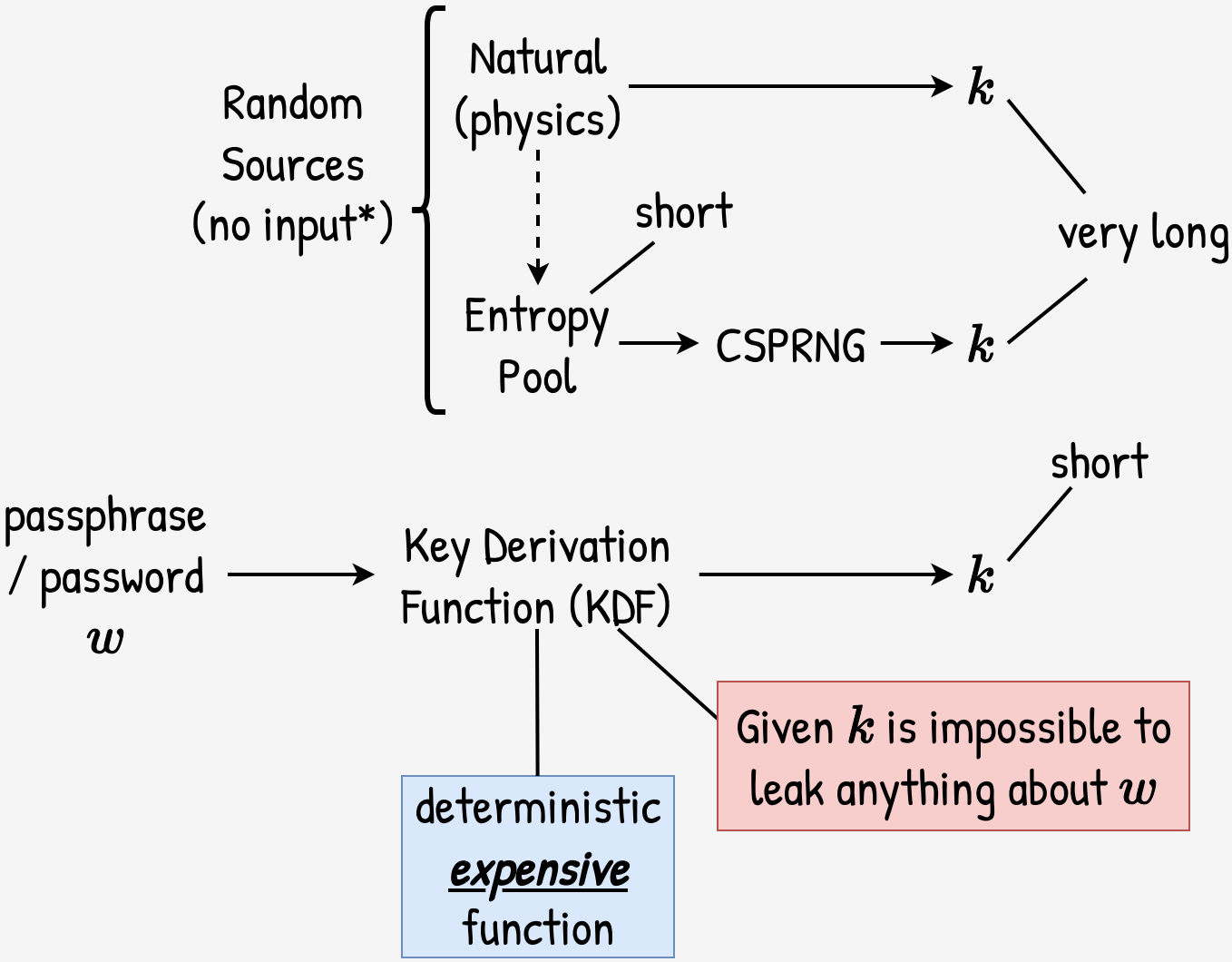
And how do we generate these?

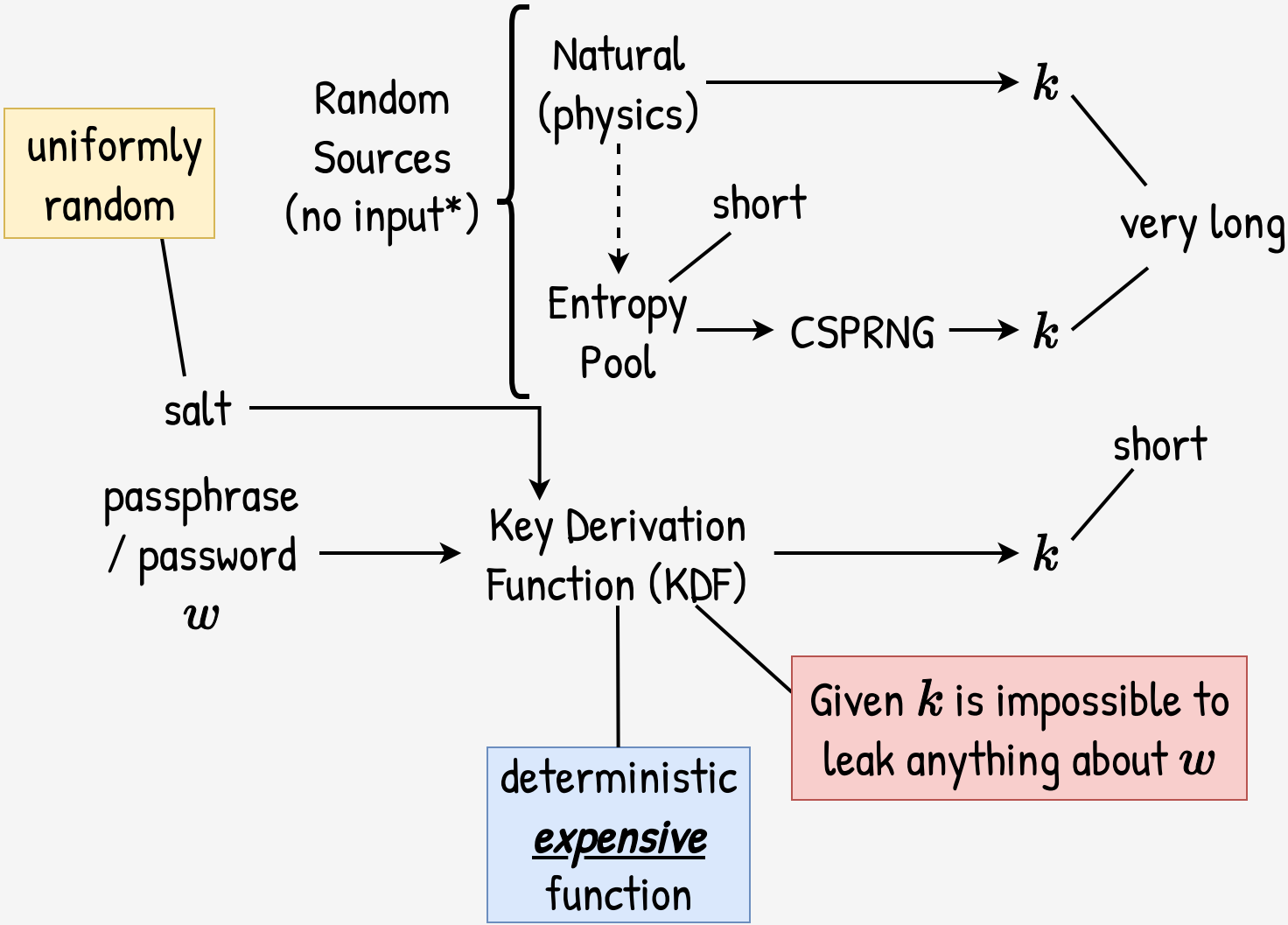
$|k| \ll |s|$

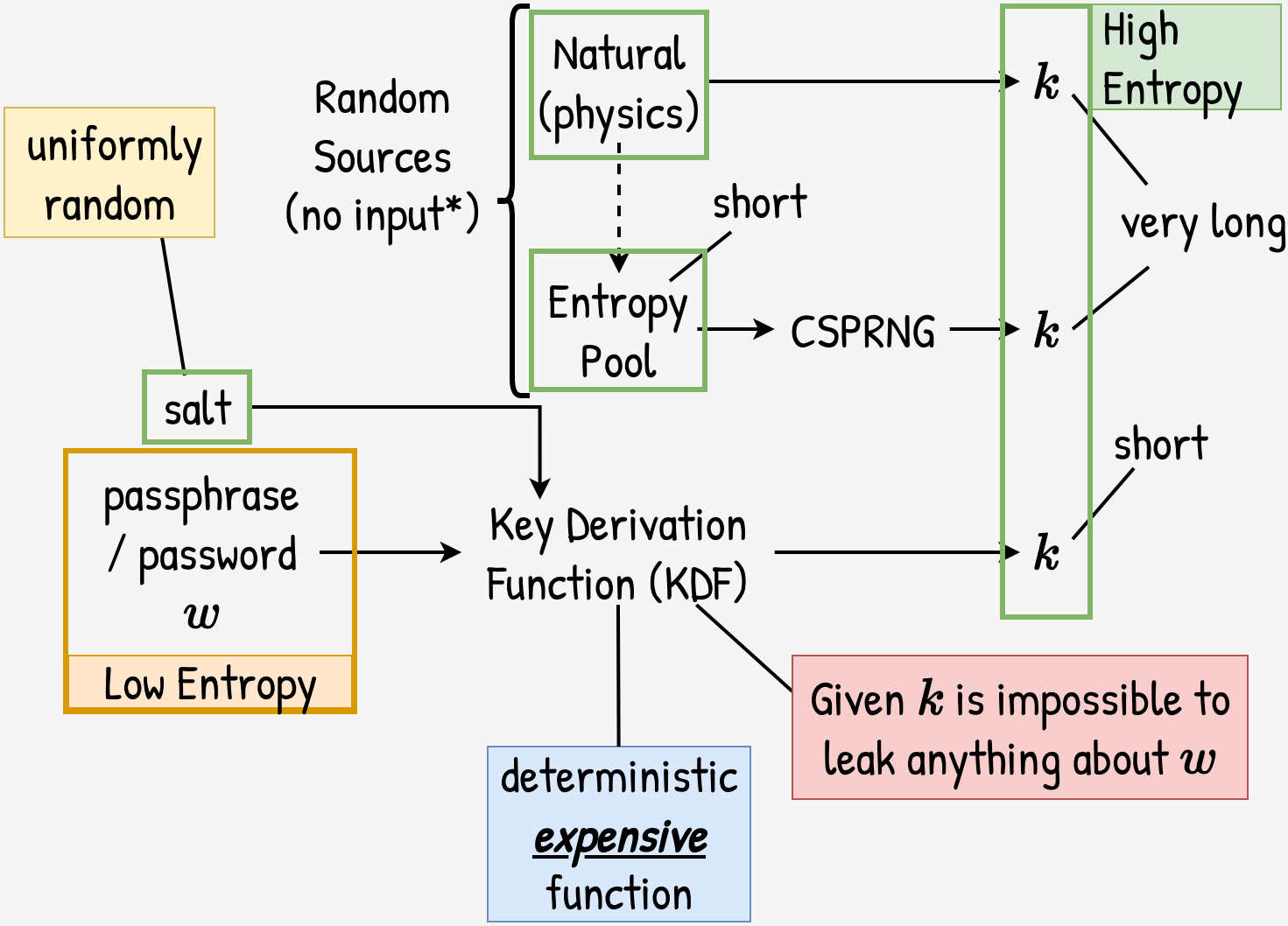
Stream Cipher

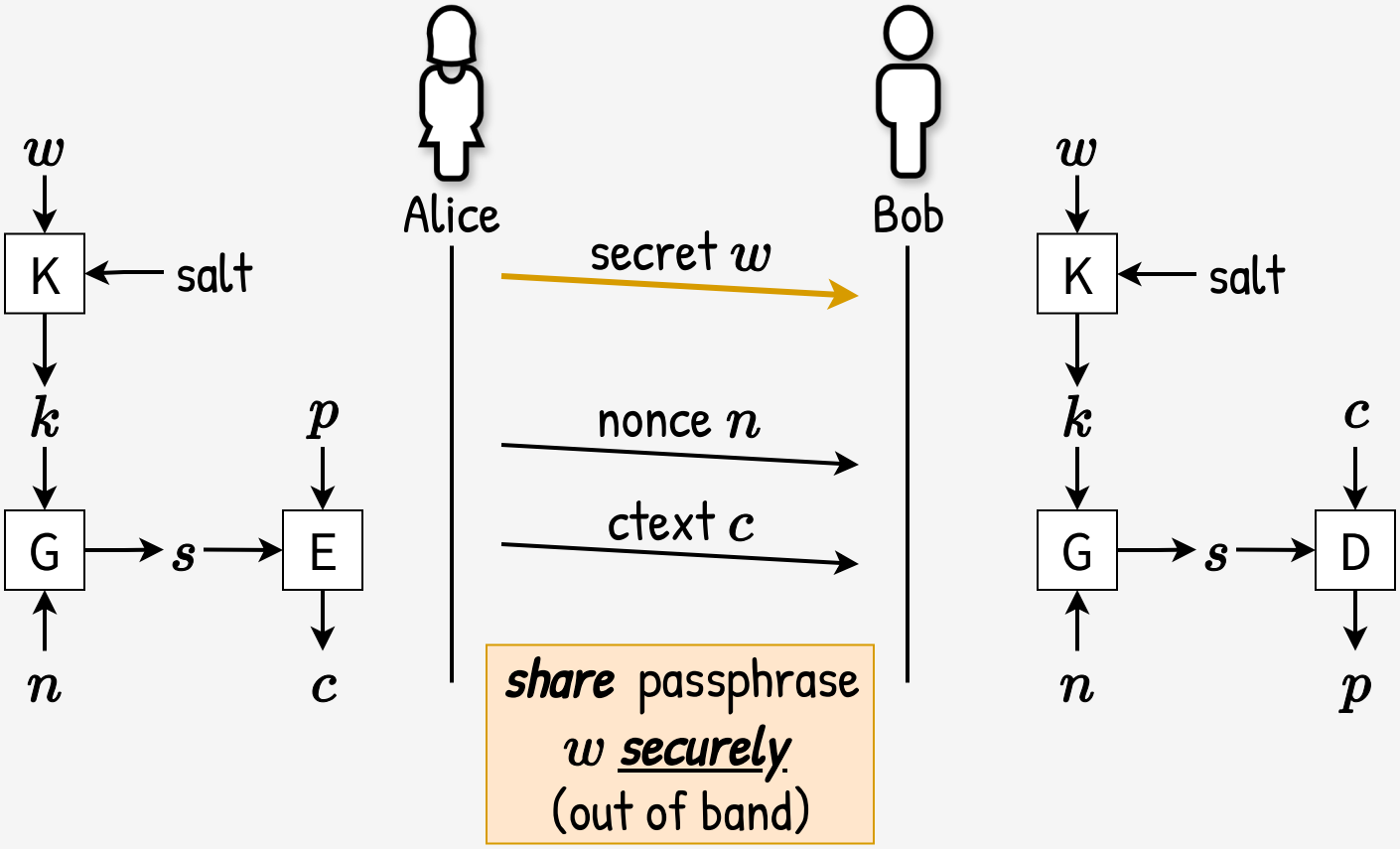

Random
Sources
(no input*)
Natural
(physics)
k
short
very long
Entropy
Pool
CSPRNG
k


Random
Sources
(no input*)
Natural
(physics)
k
short
very long
Entropy
Pool
CSPRNG
k
passphrase
/ password
w
Key Derivation
Function (KDF)
k


Random Sources (no input*)
Natural (physics)
k
short
very long
Entropy Pool
CSPRNG
k
short
passphrase / password w
Key Derivation Function (KDF)
k
Given k is impossible to leak anything about w
deterministic
expensive
function


uniformly random
Random Sources (no input*)
Natural (physics)
k
short
very long
Entropy Pool
CSPRNG
k
salt
short
passphrase / password w
Key Derivation Function (KDF)
k
Given k is impossible to leak anything about w
deterministic
expensive
function


uniformly random
Random Sources (no input*)
Natural (physics)
Entropy Pool
short
CSPRNG
k
k
k
High Entropy
very long
short
salt
passphrase / password w
Low Entropy
Key Derivation Function (KDF)
Given k is impossible to leak anything about w
deterministic ***expensive*** function



Confidenciality ✓

Cryptographic Secure
Hash Function

h should be ***uniformly distributed***

$m \longrightarrow \boxed{\text{H}} \longrightarrow h$

$$H : 2^n \rightarrow 2^k$$
$$n > k$$

deterministic function

Given h should be impossible to **recover** m

Pre-Image Resistance

It should be hard to find m_1, m_2 such $H(m_1) = H(m_2)$

Collision Resistance

Integrity ✓

Cryptographic Secure
Hash Function

h should be ***uniformly distributed***

But they are designed to be ***efficient*** so they are ***bad*** KDF

$$m \longrightarrow \boxed{\text{H}} \longrightarrow h$$

$$H : 2^n \rightarrow 2^k$$
$$n > k$$

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Pre-Image Resistance

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$m \longrightarrow \boxed{\text{H}} \longrightarrow h$

$$H : 2^n \rightarrow 2^k$$
$$n > k$$

Given h should be impossible to **recover** m

Pre-Image Resistance

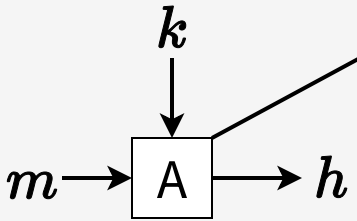
But an attacker may ***learn*** something about m

deterministic function

It should be hard to find m_1, m_2 such $H(m_1) = H(m_2)$

Collision Resistance

Integrity ✓

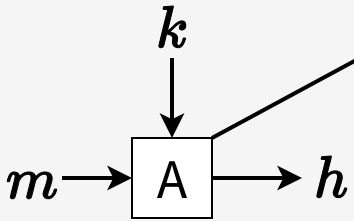


CS-Hash Function

It is really hard to find a ***valid*** (m, h) pair without the ***secret*** k

Forgery Resistant

Cryptographic Secure
Keyed Hash Function



CS-Hash Function

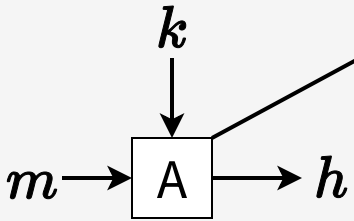
It is really hard to find a ***valid*** (m, h) pair without the ***secret*** k

Forgery Resistant

Cryptographic Secure
Keyed Hash Function

Integrity ✓

Authentication ✓



CS-Hash Function

It is really hard to find a ***valid*** (m, h) pair without the ***secret*** k

Forgery Resistant

Cryptographic Secure Keyed Hash Function

HMAC is the ***standard*** way to combine a CS-Hash with a secret k in a ***secure*** way

Integrity ✓

Authentication ✓

We can authenticate messages, but ***which*** message?

$$e = E[m]$$
$$c = e \,\|\, A[e]$$

Encrypt then Auth

$$a = A[m]$$
$$c = E[m \,\|\, a]$$

Auth then Encrypt

$$e = E[m]$$
$$a = A[m]$$
$$c = e \,\|\, a$$

$\underbrace{\text{Encrypt and Auth}}_{\text{Are all good?}}$

We can authenticate messages, but ***which*** message?

$$e = E[m]$$
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Encrypt then Auth

Secure in general if E and A are secure

$$a = A[m]$$
$$c = E[m \,\|\, a]$$

Auth then Encrypt

Secure ***but*** it may not for some E and/or A even if they are secure.

$$e = E[m]$$
$$a = A[m]$$
$$c = e \,\|\, a$$

Encrypt and Auth

Are all good?

Nope!

We can authenticate messages, but ***which*** message?

bonus: ctext unforgeable

$$e = E[m]$$
$$c = e \,\|\, A[e]$$

Encrypt then Auth

Secure in general if E and A are secure

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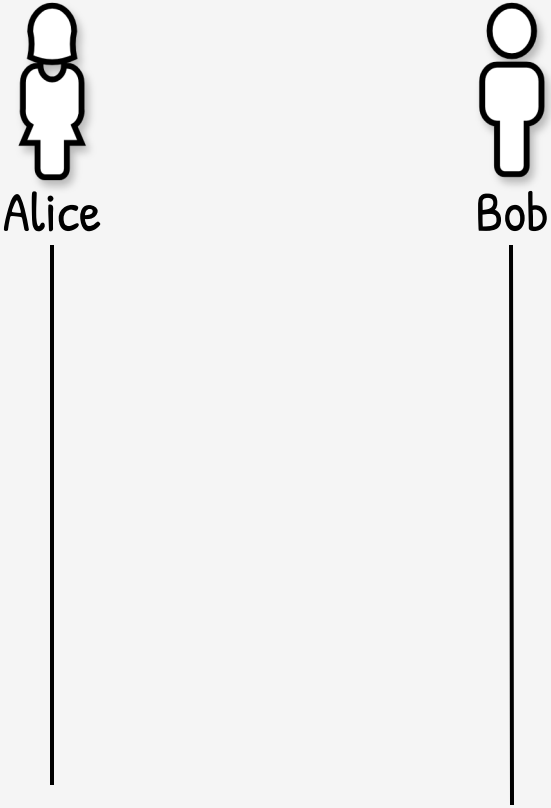
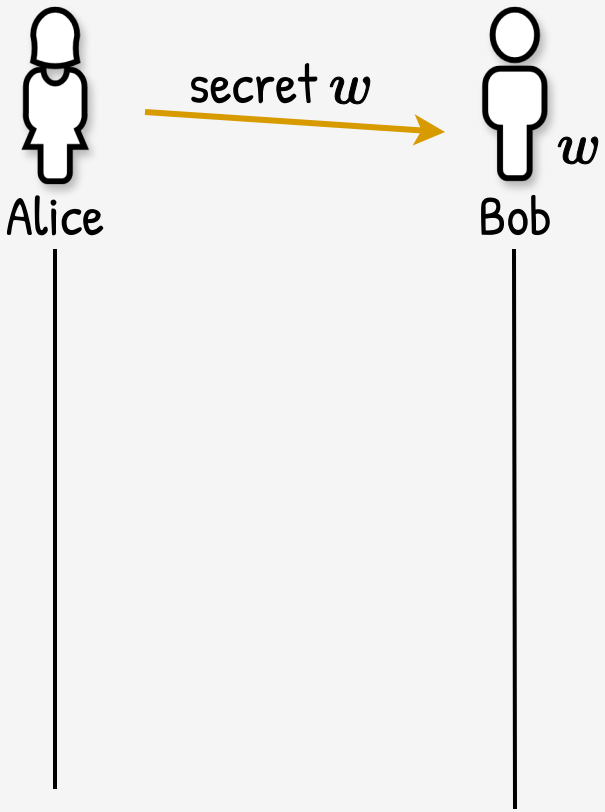
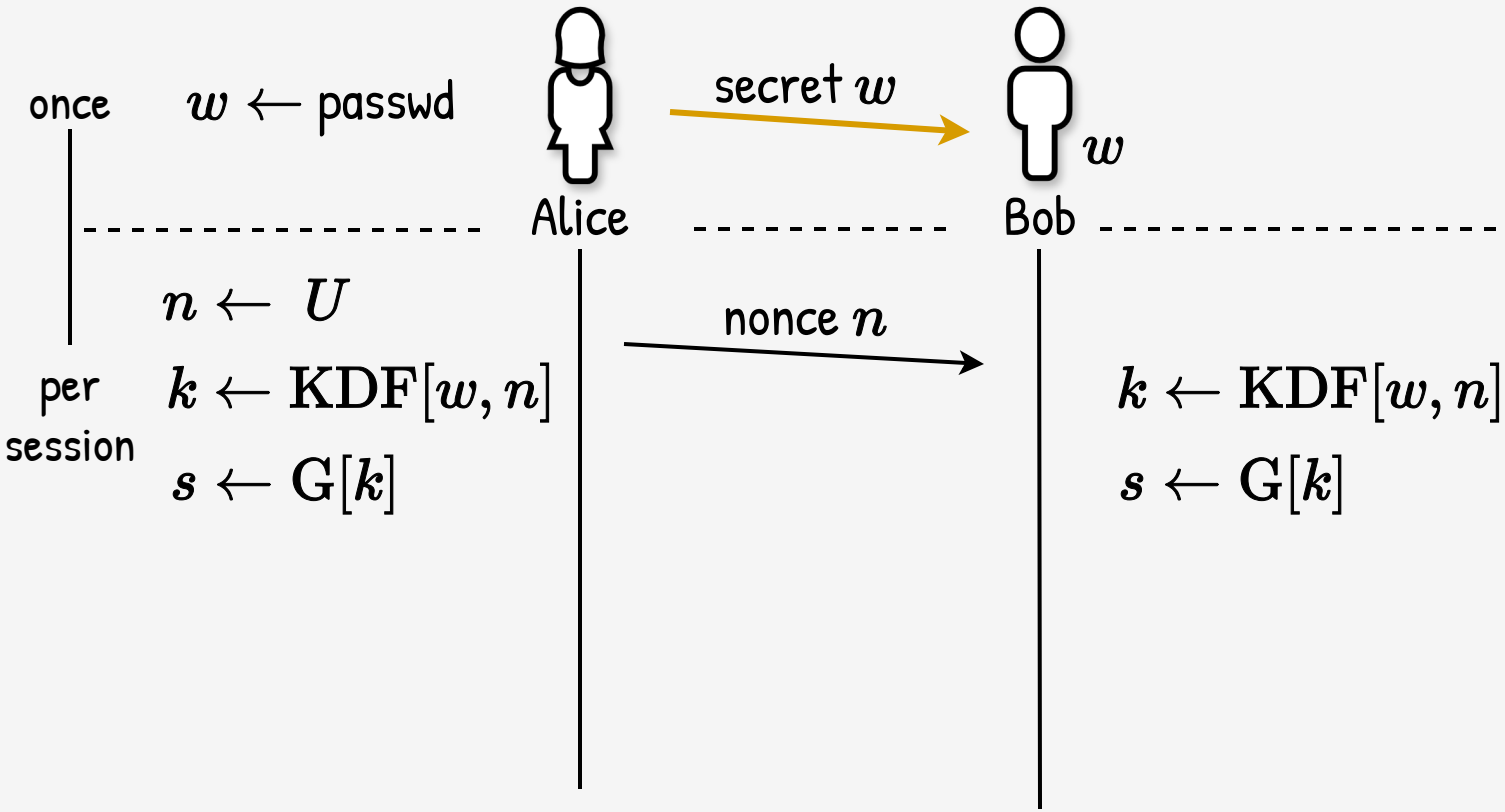
Are all good?

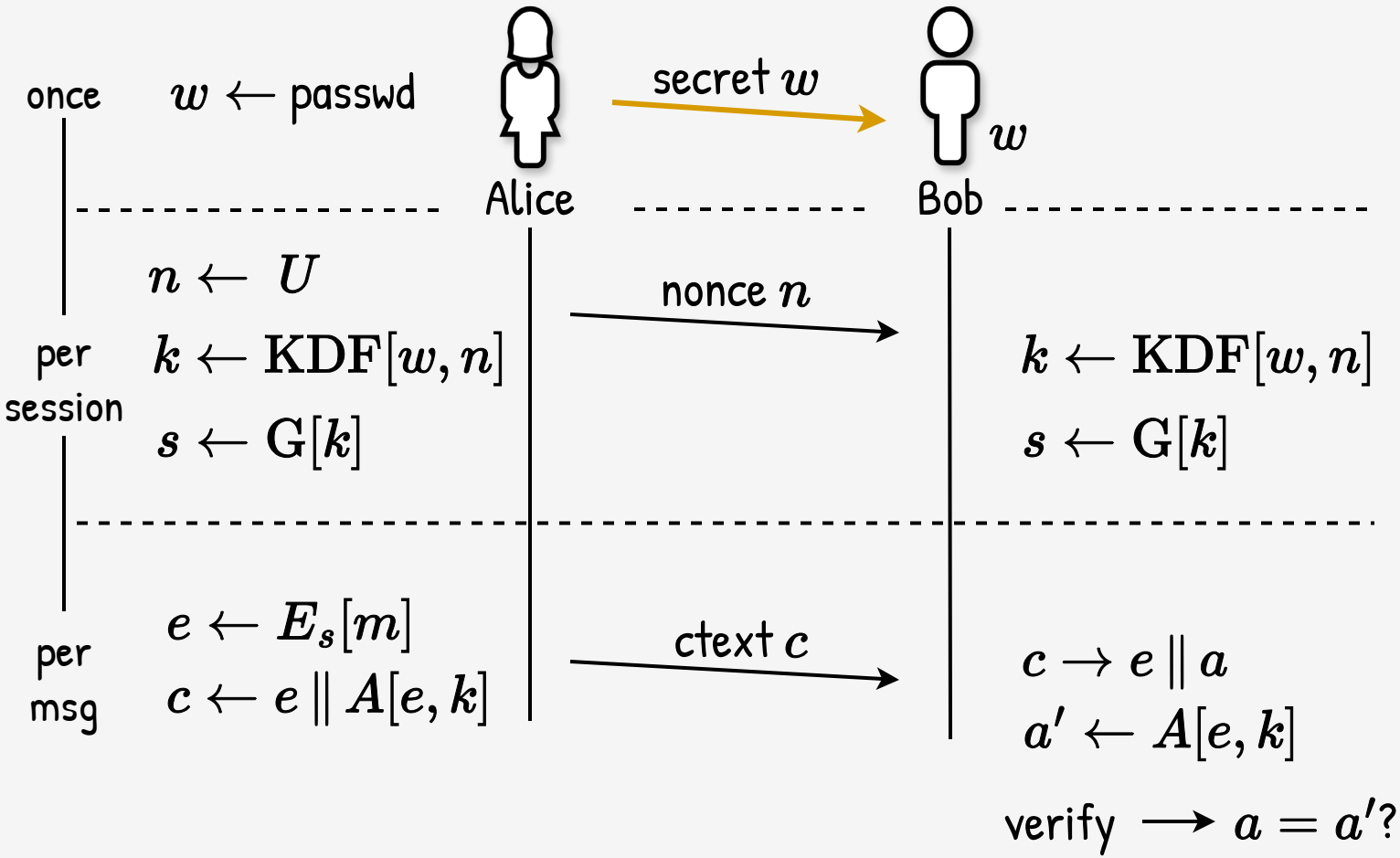
Nope!


Alice
Bob

once $w \leftarrow$ passwd




once
$w \leftarrow$ passwd
secret w
w
Alice
Bob
$n \leftarrow U$
nonce n
per session
$k \leftarrow \mathrm{KDF}[w, n]$
$s \leftarrow \mathrm{G}[k]$
$k \leftarrow \mathrm{KDF}[w, n]$
$s \leftarrow \mathrm{G}[k]$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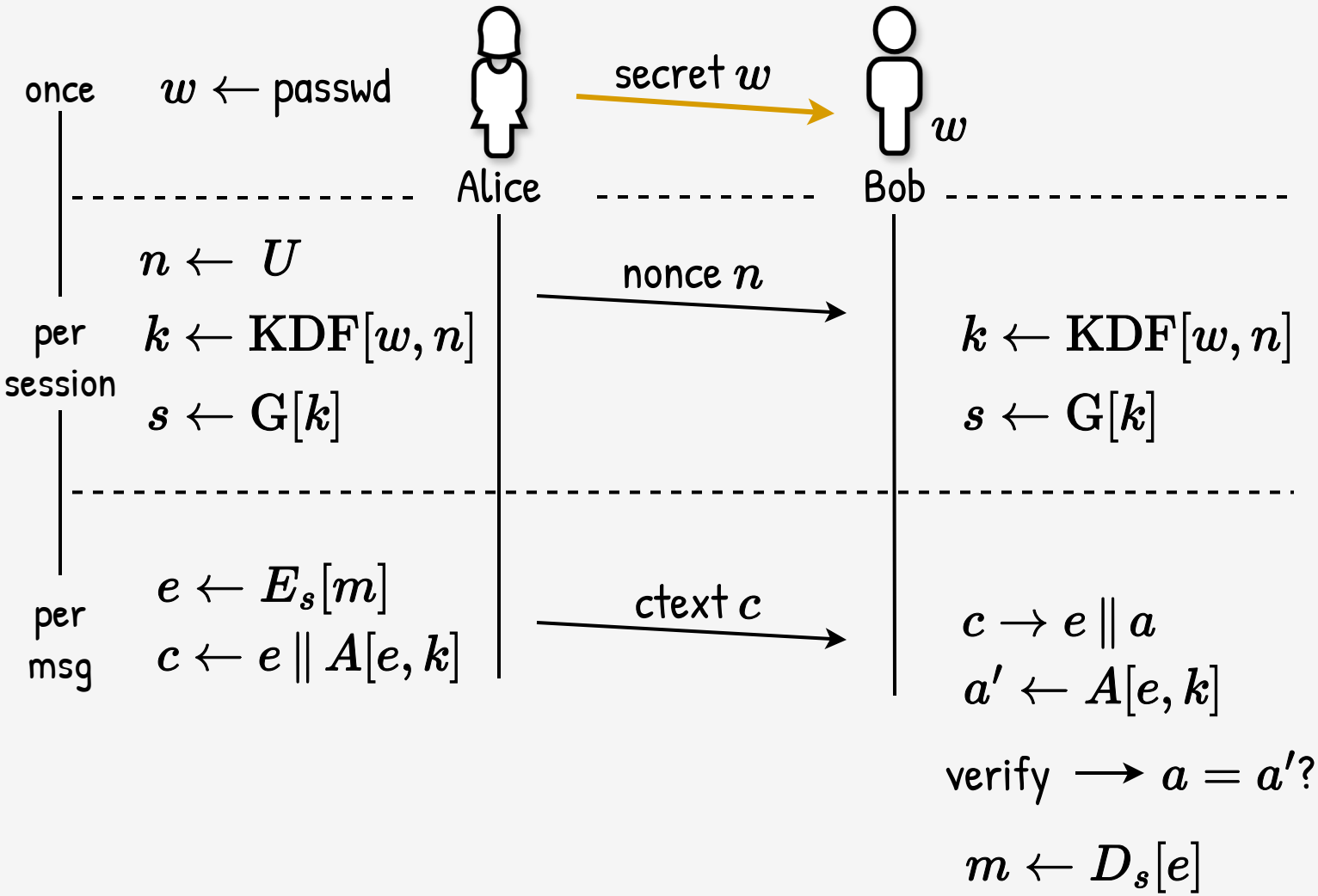
nonce n
$k \leftarrow \mathrm{KDF}[w, n]$
$s \leftarrow \mathrm{G}[k]$
per msg
$e \leftarrow E_s[m]$
$c \leftarrow e \,\|\, A[e, k]$
ctext c
$c \rightarrow e \,\|\, a$
$a' \leftarrow A[e, k]$
verify $\longrightarrow a = a'$?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$w \leftarrow$ passwd
secret w
w
Alice
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$n \leftarrow U$
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$s \leftarrow \mathrm{G}[k]$
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$e \leftarrow E_s[m]$
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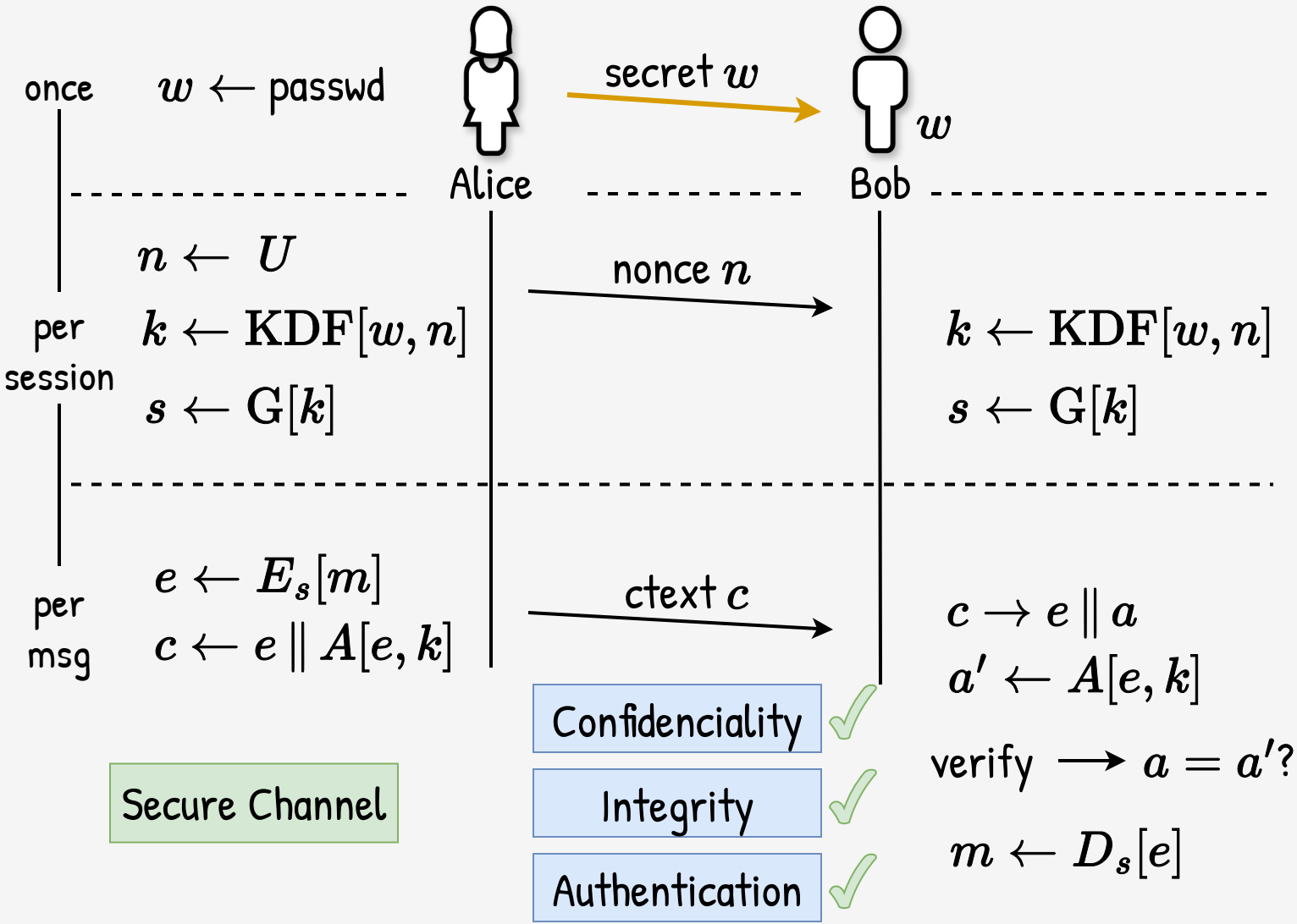
verify $\longrightarrow a = a'$?
$m \leftarrow D_s[e]$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secret w
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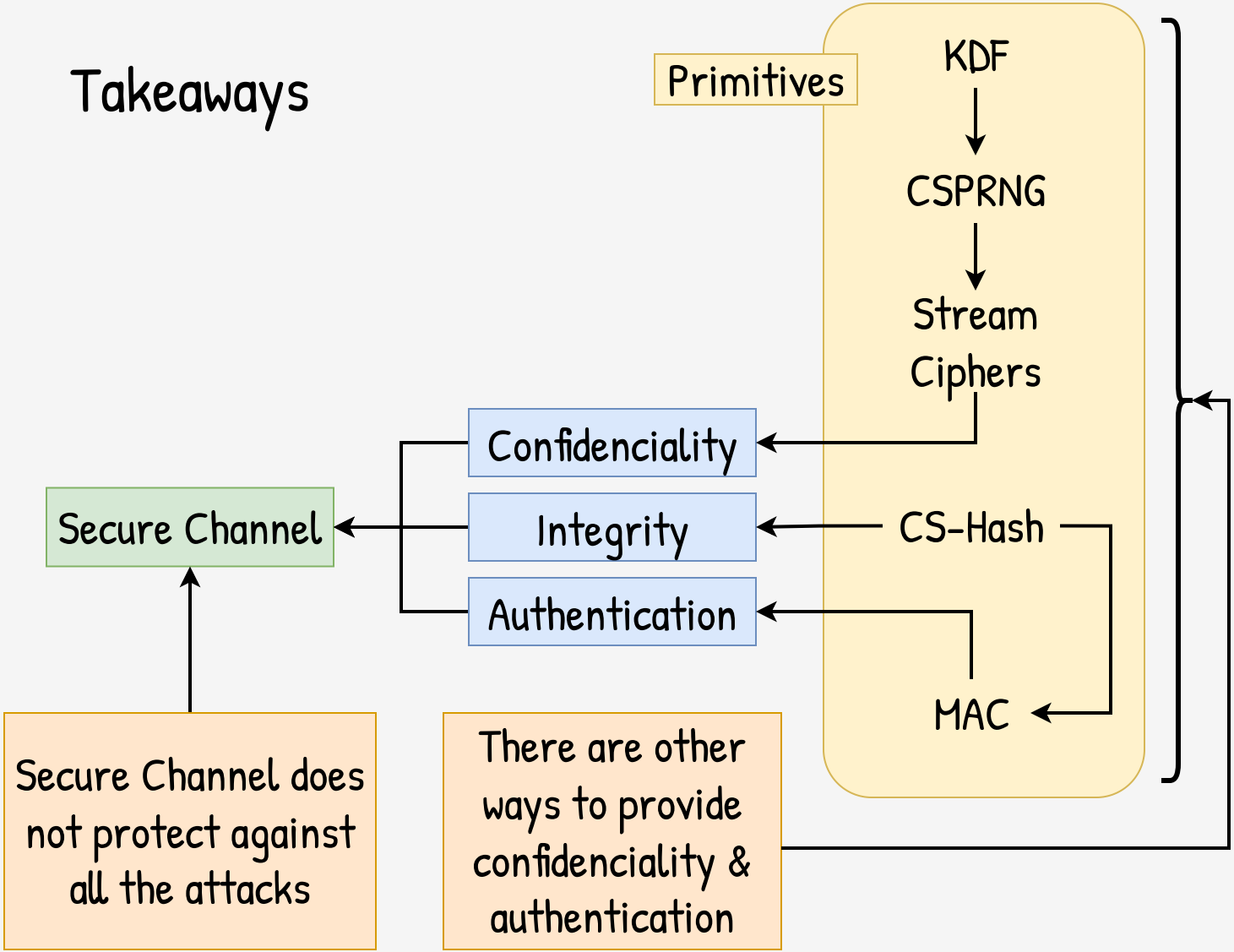
$n \leftarrow U$
$k \leftarrow \mathrm{KDF}[w, n]$
$s \leftarrow \mathrm{G}[k]$
nonce n
$k \leftarrow \mathrm{KDF}[w, n]$
$s \leftarrow \mathrm{G}[k]$
per msg
$e \leftarrow E_s[m]$
$c \leftarrow e \,\|\, A[e, k]$
ctext c
$c \rightarrow e \,\|\, a$
$a' \leftarrow A[e, k]$
verify $\longrightarrow a = a'$?
$m \leftarrow D_s[e]$
Confidenciality
Integrity
Authentication
Secure Channel

Takeaways


Primitives
KDF
CSPRNG
Stream Ciphers
CS-Hash
MAC
Confidenciality
Integrity
Authentication
Secure Channel
Secure Channel does not protect against all the attacks
There are other ways to provide confidenciality & authentication